

Tharun Balaji S

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EDUCATION

PSG Institute of Technology and Applied Research

B.E in Mechanical Engineering | CGPA: 8.23

Expected 2027

Coimbatore, India

Lisieux Matric Hr Sec School

12th HSE – 87.83%

2021 – 2023

Coimbatore, India

WORK EXPERIENCE

Backend Software Engineer

Bluecode Solution

Dec 2025 – Present

- Developed and integrated REST APIs for Apartment Management SaaS modules using Node.js, Fastify, and PostgreSQL, improving feature scalability and system maintainability.
- Implemented modular and decoupled backend architecture with collaborative Git workflows, reducing integration complexity and enabling parallel development across multiple teams.

Data Analyst

Outrix

Nov 2025 – Dec 2025

- Engineered a fraud detection analytics platform processing 600k+ records using Python and Pandas, achieving 88% anomaly detection precision and reducing operational costs by 45%.
- Designed merchant risk scoring dashboards with statistical outlier detection algorithms, reducing false positive alerts by 25% and accelerating analyst triage by 2x.

Machine Learning Engineer

Appin Technology

Feb 2025 – Mar 2025

- Designed and trained 5+ machine learning models using Python (Scikit-learn, Pandas, NumPy), achieving up to 85% accuracy and improving predictive performance across multiple tasks.
- Built scalable data preprocessing and EDA workflows using Pandas, Matplotlib, and Seaborn, reducing preprocessing time by 40% and accelerating model development cycles.

PROJECTS

CivicFlow – AI Driven Civic Intelligence

- Built an AI-driven civic reporting system leveraging RAG pipelines and duplicate detection techniques to enhance validation efficiency and optimize issue processing workflows.
- Implemented multimodal computer vision and LLM solutions using CLIP, YOLOv8, and PRNU for image verification, intelligent classification, and automated routing of civic reports.

EDUOS – College Management SaaS Tool

- Developed an AI-powered college management SaaS platform using React, Node.js, and Firebase with features including automated attendance tracking, marks management, and AI-assisted resume generation.
- Built scalable RESTful APIs and role-based dashboards for students, faculty, and administrators with real-time analytics and integrated AI chatbot support.

Cluster Analysis - Based Fall Detection Using Wearable Device

- Designed and developed an intelligent wearable monitoring system using sensor fusion techniques by integrating accelerometer and heart rate sensor data for real-time fall detection and physiological monitoring.
- Built cluster analysis-based activity classification models using Python and Scikit-learn to identify abnormal movement patterns and improve fall detection accuracy.

ACHIEVEMENTS

- **Amazon ML Challenge 2025** - Processed and analyzed large-scale datasets in the Amazon ML Challenge 2025, against 2000+ teams and achieving **Runner-Up** position through effective ML model and optimization.
- **Ideathon 2024 - Finalist** - Developed an Aadhaar-based facial recognition system for secure identity verification and authentication, achieving Top 10 finalist status among participating teams in the college competition.
- **APTIV Abilities Accessibility XIdeathon 2025 - Finalist** - AI-driven civic intelligence solution for smart issue validation and accessibility enhancement.
- **Coding Club Executive Member** - Organizing and coordinating coding events, hackathons, and technical workshops

TECHNICAL SKILLS

- **Languages:** Python, C++, SQL
- **AI & ML:** Scikit-learn, TensorFlow, Pandas, NumPy, Seaborn, RAG Pipelines, CLIP, YOLOv8.
- **Frameworks:** FastAPI, React, Node.js, Express, Fastify
- **Databases:** PostgreSQL, MongoDB, MySQL, Redis
- **Tools:** Git, GitHub, OpenCV, Postman, Docker, Jupyter Notebook, Vercel
- **Concepts :** Operating System(OS), Computer Networks, OOPS